DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
HALF YEARLY EXAMINATION

CLASS: X | **FULL MARKS: 80**
SUBJECT: BIOLOGY [SET A] | **TIME: 2 HOURS**

Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.

Attempt all questions from Section I and any four questions from Section II.
The intended marks for questions or parts of questions are given in brackets [ ].
This question paper consists of 6 printed pages.

SECTION A (40 Marks)
(Attempt all questions from this Section)

Question 1.

Select the correct answers to the questions from the given options. (Do not copy the question. Write the correct answer only): [15]

i. Which sex chromosomes are present in all mature human sperm cells?
- a. Both X and Y
- b. Either X or Y
- c. Only X
- d. Only Y

ii. Sana went to the lab with her biology teacher and was observing slides with different stages of Mitosis. In one of the slide, she observed that the chromosomes are lined up in one plane at the equator. Which stage has been referred to in the below given options?
- a. Prophase
- b. Metaphase
- c. Anaphase
- d. Telophase

iii. The space between the plasma membrane and the cell wall in a plasmolysed cell is filled with:
- a. Isotonic solution
- b. Hypotonic solution
- c. Hypertonic solution
- d. Water

iv. Root hair is:
- a. Extension of cortex
- b. Extension of endothelium
- c. Extension of epidermal cells
- d. Extension of endodermis

v. Ravi has a beautiful garden and he tried his hands in crossing a pure tall, round seed plant and a pure dwarf, wrinkled seed plant to get new varieties in the offsprings. What will be the phenotypic ratio of F_2 offsprings?
- a. 1:2:1
- b. 9:3:3:1
- c. 3:1
- d. 9:2:3:1

vi. **Hormone which controls the production of urine in the kidney is**

a. ADH **b. ACTH**

c. FSH **d. TSH**

vii. **If a pure purple flower plant (AA) is crossed with a pure white flower plant (aa), the genotype of the offsprings of F_1 generation will be:**

a. All AA **b. All Aa**

c. 3 AA and 1 aa **d. 1AA: 2 Aa: 1aa**

viii. **Which of the following might be the backbone of DNA?**

1. Phosphate, 2. Sugar, 3. Adenine

a. 1, 2 and 3 **b. Only 2**

c. 1 and 2 **d. 2 and 3**

ix. **The rate of transpiration increases with one or more external factor/s. Which of the following external factor/s affects the increase of transpiration?**

1. Increase in humidity **2. Increase in wind velocity**

3. Reduced light intensity **4. Reduced water availability**

a. Both 1 and 2 **b. Only 2**

c. Only 3 **d. Both 2 and 3**

x. **Internodal elongation is associated with:**

a. Auxin **b. Cytokinin**

c. Abscisic acid **d. Gibberellins**

xi. **The respiratory pigment which is present in red blood cells is:**

a. Haemoglobin **b. Anthocyanin**

c. Chlorophyll **d. Melanin**

xii. **The site of maturation of human sperm is:**

a. Epididymis **b. Interstitial cells**

c. Prostate gland **d. Seminiferous tubules**

xiii. **Heart is covered by a membranous covering called:**

a. Myocardium **b. Pericardium**

c. Epicardium **d. Endocardium**

xiv. **Phytohormones which help in germination of seeds is:**

a. ABA **b. Auxin**

c. IAA **d. Cytokinin**

xv. **Assertion (A) - Right side of the human heart has deoxygenated blood. Reason (R) – Vena cava is the artery that supplies deoxygenated blood to the heart.**

a. Both A and R are true **b. A is true and R is false**

c. A is false and R is true **d. Both A and R are false**

Question 2

(i) Name the following: [5]

- **a. The onset of menstruation in young girls at the age of 13 years.**
- **b. The innermost layer of eye where image is formed.**
- **c. The smallest blood vessel formed by union of capillaries.**
- **d. The tropic movement of plants in response to light.**
- **e. The suppressed allele of a gene.**

(ii) Choose the odd one out and write the correct category to which the others belong: [5]

- **a. Pons, Corpus luteum, Corpus callosum, Cerebellum.**
- **b. Vas deferens, Fallopian tube, Epididymis, Cowper's gland.**
- **c. GH, TSH, Vasopressin, LH.**
- **d. Incus, Malleus, Pinna, Stapes.**
- **e. RBC, ATP, WBC, Platelets.**

(iii) Fill in the blanks with suitable words: [5]

Water enters the root hairs from the soil by the process of ___a_____. This is because the solution in the soil is ___b____. The water molecules travel through ____c____ upward in the plants from the roots to the leaves. The process by which sap oozes out of the injured part of the plant is known as ____d________. The pressure exerted by the cell wall on the cell content is known as ___e_____.

(iv) Arrange and rewrite the terms in each group in the correct order so as to be in a logical sequence beginning with the term that is underlined. [5]

- **a. Uterus, Parturition, <u>Fertilisation</u>, Implantation.**
- **b. Auditory canal, Ear ossicle, Cochlea, <u>Pinna.</u>**
- **c. Bowman's capsule, Loop of Henle, <u>Afferent arteriole</u>, DCT.**
- **d. Caterpillar, Snake, Frog, <u>Leaves</u>.**
- **e. Cortex, <u>Root hair</u>, Endodermis, Xylem.**

(v) Match the items given in Column I with the most appropriate ones in Column II and rewrite the correct matching pairs. [5]

Column I	**Column II**
a. Movement of water	**a. Cell division**
b. Grana	**b. Carry blood to lungs**
c. Cytokinin	**c. Xylem**
d. Tympanic membrane	**d. Light reaction of photosynthesis**
e. Coronary artery	**e. Static Equilibrium**
	f. Ear drum
	g. Carry oxygenated blood to the heart muscle

SECTION B (40 Marks)

(*Attempt any four questions from this section)*

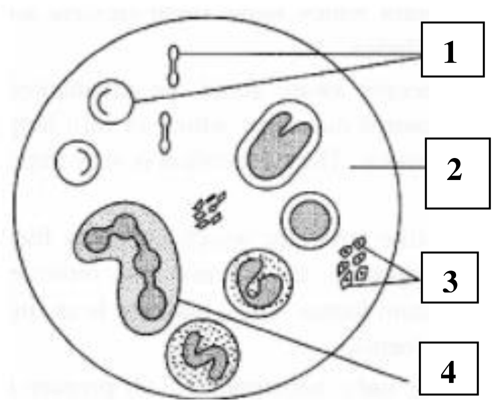
Question 3

(i) Define Guttation. **[1]**

(ii) Mention the two functions of spinal cord. **[2]**

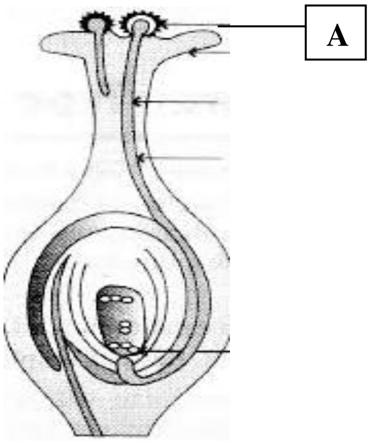
(iii) What is meant by power of accommodation of the eye? Name the muscle of the eye which is responsible for the same. **[2]**

(iv) Observe the diagram and answer the following questions: **[2]**



Name the soluble protein found in '2'. Nucleus is absent in matured structures of '1'. Explain with proper reasons.

(v) Observe the diagram given below and answer the following questions: **[3]**



a. Name the type of tropism shown in the diagram.
b. Define the above mentioned tropism from question 'a'.
c. Give an example of a stimulant that helps in the growth of the part marked 'A'.

Question 4

(i) Mention the balanced chemical equation of photosynthesis. **[1]**

(ii) Name the organ where semen is produced. Mention the function of Inguinal canal. **[2]**

(iii) What type of reflexes are the following? **[2]**
a. Knitting
b. Blinking of the eye

(iv) On sprinkling excess of fertilizers in the field the plants wilt. Justify with reasons. [2]

(v) A homozygous plant with purple flower (R) is crossed with homozygous plant with white flower (r) [3]

a. Mention the Genotype of F_1 generation.

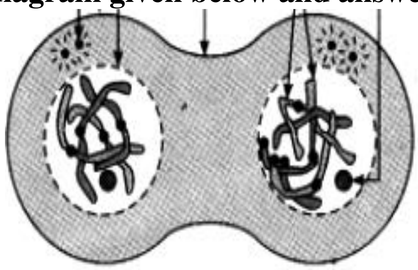
b. Explain the Mendel's Law which can be associated with this cross.

c. Give the phenotypic ratio of F_2 generation.

Question 5

(i) Define osmosis. [1]

(ii) Observe the diagram given below and answer the following questions: [2]



a. Identify whether it is a plant cell or an animal cell with proper reasons to support your answer.

b. Mention any one main feature of the stage which comes before this stage.

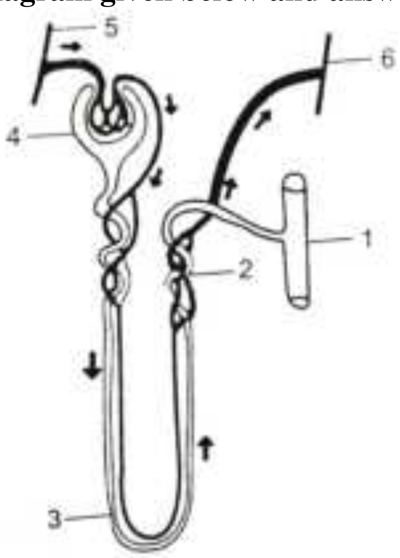
(iii) "Matured Red Blood Cells in human being lack mitochondrion." Explain with reasons. [2]

(iv) On a bright sunny day water weeds growing in an aquarium were actively giving off bubbles of gas. Use this information and answer the following questions: [2]

a. Name and define the process occurring in water weeds that has resulted in evolution of these bubbles.

b. Which gas is present in these bubbles?

(v) Study the diagram given below and answer the following questions: [3]



a. Name the blood vessel which contains least amount of urea in this diagram.

b. What type of blood is present in "5"?

c. Mention the function of '3'.

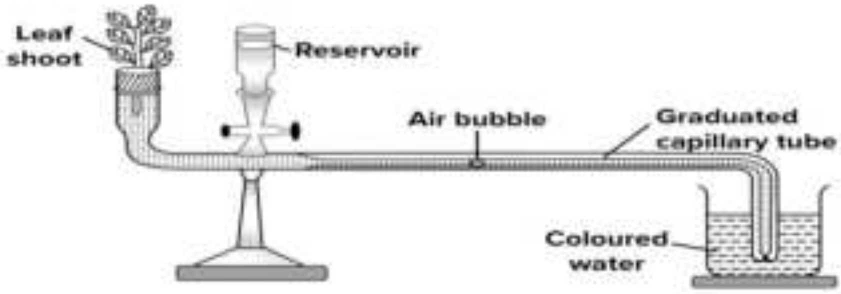
Question 6

(i) Name the gland which secretes the hormone to regulate the normal growth of a person. [1]

(ii) What do you understand by the term "double circulation" of blood in human? [2]

(iii) "Shrubs are pruned for a better bushy growth". Explain with reasons. [2]

(iv) Observe the diagram given below and answer with reasons: [2]



a. Identify the apparatus.

b. Why do we use coloured water during the experiment?

(v) Draw a neat labelled diagram of cross section of an artery of human body. [3]

Question 7

(i) Mention the name of the paper which is used as an indicator of moisture. [1]

(ii) Define Karyotype. Name the inherited disease due to which an individual cannot differentiate between certain colours. [2]

(iii) Mention a neurotransmitter present in the synapse. Define nephron. [2]

(iv) A thin strip of epidermal cells from the fleshy scales of an onion bulb was taken and transferred to strong sugar solution. On the basis of this answer the following questions: [2]

a. Mention the state of the cell observed after sometime when it was placed in strong solution.

b. What would you do to bring the cell back to its original condition?

(v) Draw a neat labelled diagram of the plant cell which you can sketch by reading the lines given in question (iv). [3]

Question 8

(i) Mention the full form of IAA. [1]

(ii) Mention the scientific name of the plant used by Mendel to conduct his experiments. Mention one reason to choose that particular plant for his experiments. [2]

(iii) Explain with reasons: [2]
Fewer stomata and narrow leaves affect transpiration in a plant.

(iv) Mention the functions of Uterus and Seminiferous tubule. [2]

(v) Draw a neat labelled diagram of a chloroplast. [3]